Test Report No. 2024 – 0452

Agricultural Machinery – Feed Pellet Mill
DAVAO TECHNOCRAFT
GMPMVT1024

College of Engineering and Agro-Industrial Technology, University of the Philippines Los Baños, College, Laguna 4031
Email: amtec.uplb@up.edu.ph   Facebook: www.facebook.com/amtec.uplb   Website: www.amtec.ceat.uplb.edu.ph
Contact Numbers: +63 921 400 7137 (Smart) / +63 917 704 0922 (Globe)

Test Report Number	2024 – 0452
Date of Issuance	November 26, 2024
Valid Until	November 26, 2029
Agricultural Machinery	Feed Pellet Mill
Type	Flat die pellet mill, Die-rotating
Brand	DAVAO TECHNOCRAFT
Model	GMPMVT1024
Test Requested by	DAVAO TECHNO CRAFT 424 Echavez St., P-4 Poblacion Nabunturan, Davao de Oro

SUMMARY

Performance Criteria	Requirement as per PNS/BAFS PABES 274:2019	AMTEC Test
Pelleting capacity, kg/h	200[a]	452.75
Pelleting efficiency, %, minimum	92	92.22
Pelleting recovery, %, minimum	95	96.88
Pellet diameter's coefficient of variation, %, maximum	7	2.24

[a] Data obtained from the operator's manual provided by the applicant.
Note: ND – No Data

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Not provided
e. Maintenance instruction	Provided
f. Storage	Provided
g. Handling, reception, transportation, assembly, and installation	Provided
h. Specifications	Provided
i. Dismantling and disposal	Provided
j. Warranty	Provided
k. Parts list	Provided
2. Previous test report of similar brand and model	None
3. Manufacturer	DAVAO TECHNO CRAFT
4. Serial number	PMELEC050624
5. Prime mover used	3.7 kW MINDONG YC132SB-4 Electric Motor
6. Number of operators	Two (One feeding/machine operator and bagger/assistant)
7. Loading of input	The feed mix were manually loaded in the input hopper. The hopper is accessible to the operator.
8. Cleaning of parts	Parts were accessible for cleaning by dismantling or removing its covers.
9. Adjusting of parts	a) Feed intake rate may be adjusted through the sliding gates provided at the input hopper. b) Clearance of the roller and die can be adjusted thru the tightening bolts. The length of pellet can also be adjusted using the bolts provided. c) Belt and transmission tension can also be adjusted by moving the prime mover along its base and by adjusting tensioner roller provided.
10. Collecting output	The pellets were collected by steel bins and was transferred into a sack.
11. Safety features	a) On/Off buttons were provided b) All rotating or moving parts are covered.

OBSERVATIONS (Cont'n)

12. Failure or abnormalities that may be observed on the machine or its component parts during and after the operation	No failure or abnormalities were observed on the machine or its components during and after the operation.
13. The hopper shall be made from any corrosion-resistant and smooth material to reduce feed mix from sticking to the inside surface of the hopper and for ease in cleaning. Auxiliary devices should be provided to reduce sticking or bridging of feed mix to the surface of the hopper.	The hopper was made of 303/304SS as determined using the BRUKER S1 TITAN 500 X-ray Fluorescence Analyzer. See Table 1 for the elemental composition of the materials. There was no sticking of feed mix on the surface of the hopper during the test. No auxiliary devices to reduce sticking or bridging of feed mix was provided on the hopper; manual removing or mixing of feed mix on the hopper was observed during the operation. The hopper is removable and is accessible for cleaning.
14. Pelletizing components of the machine such as conditioner, pelleting chamber, roller, barrel and screw, and die shall be made from corrosion and wear-resistant materials.	Pelletizing components of the machine such as die and roller are made of 303/304SS, while the pelleting chamber is made of 100 Series, as determined using the BRUKER S1 TITAN 500 X-ray Fluorescence Analyzer. See Table 1 for the elemental composition of the materials.
15. The pelleting chamber shall be able to withstand the force developed by the rotating die and/or rollers during operation.	No deformation or crack was observed on the pelleting chamber during and after the test operation.
16. Die and roller should be heat treated to prevent premature breakage.	Heat treatment on the die and roller was not verified due to lack of technical capability on-site.
17. Steel bars (angle bars and flat bars), metal sheet or plate, mild steel, stainless steel, and other steel materials shall be generally used in fabricating the other components of the pellet mill.	Other components of the machine such as hopper and discharge chute are made of 303/304SS, while the main frame is made of Iron/Carbon Steel, as determined using the BRUKER S1 TITAN 500 X-ray Fluorescence Analyzer. See Table 1 for the elemental composition of the materials.
18. Bolts and nuts, screws, bearings, bushing and seals to be used shall conform to PAES or other international standards.	The conformity of bolts and nuts, screws, bearings, bushing and seals used to the requirements of PAES or other international standards was not verified due to the limited technical capabilities on determining its physical and mechanical properties on site.

OBSERVATIONS (Cont'n)

19. The noise level should conform with the provisions given in Annex B of PNS/BAFS PABES 274:2019.	The maximum noise level during with load for the feeding operator and bagger are 86.1 dB(A) and 86.2 dB(A), respectively, measured at the assistant/bagger's ear level.
20. The feed pellet mill shall be free from any manufacturing defects that may be detrimental to its operation.	The feed pellet mill is free from any manufacturing defects that may be detrimental to its operation.
21. The base or frame assembly of the feed pellet mill shall be rigid and durable without any noticeable cracks and weak joints.	No noticeable cracks and weak joints on the base or frame assembly were observed.
22. The rotating components of the feed pellet mill shall be statically and dynamically balanced for stable running.	The dynamic and static balance of the rotating parts were not verified due to unavailability of test equipment to measure the parameter on site.
23. The pelleting mechanism shall be adjustable and replaceable when needed.	a) The clearance between the rollers and die is adjustable. b) Adjustment for the pellet length was also provided. c) Locally fabricated parts are repairable ang replaceable.
24. There shall be provision for adjusting the clearance between the die and the rollers.	Clearance of the roller and die can be adjusted thru the tightening bolts.
25. All metal surfaces shall be free from rust.	No rusts were observed on the machine.
26. Mechanism for emergency stop shall be provided.	On/off buttons are provided for emergency stop of operation.
27. All moving parts, sharp edges and rough surfaces shall be provided with safety features. Warning notices shall be provided in accordance with PNS/BAFS 330:2022.	All moving and rotating parts were provided with covers. Warning notices were also provided on the machine.
28. There shall be provision for access to parts during repair, maintenance, and operation.	All parts are accessible for repair and maintenance by disassembling.
29. There shall be provision for belt tightening and adjustments.	Belt and transmission tension can also be adjusted by moving the prime mover along its base and by adjusting the tensioner roller provided.
30. Other remarks	a) Two extra pellet dies with different hole diameter were provided by the test applicant (Figure 7). b) The operators were provided with earmuffs during the test operation. c) Safety and warning notices are provided on the machine.

PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing
Date of Test	September 19, 2024
Location of Test	Prk.4, Brgy. Poblacion, Nabunturan, Davao de Oro 7° 36' 20.22" N, 125° 57' 55.52" E
Items Tested	Operator's Manual, Pelleting Capacity, Pelleting Efficiency, Pelleting Recovery, Output Quality, Speed of Components, Power Consumption, Noise Level, and Number of Operators

METHODS OF TEST

The feed pellet mill was tested based on the PNS/BAFS PABES 275:2019 Agricultural Machinery – Feed Pellet Mill – Methods of Test.

DESCRIPTION OF THE MACHINE

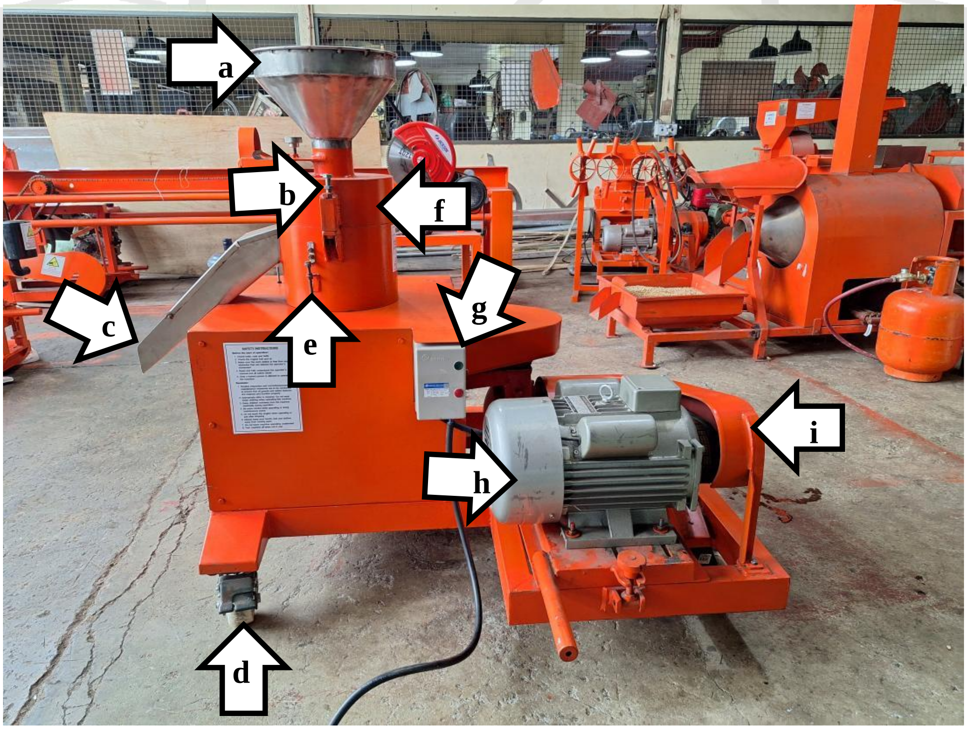

Figure 1. Parts of the DAVAO TECHNOCRAFT GMPMVT1024 Feed Pellet Mill powered by 3.7 kW MINDONG YC132SB-4 Electric Motor include the following: (a) Input hopper, (b) Adjustment bolts for clearance of die and rollers, (c) Output chute, (d) Caster wheels, (e) Adjustment bolt for pellet lengths, (f) Pelleting chamber, (g) Control panel, (h) Prime mover, and (i) Transmission system with cover.

SPECIFICATIONS

Item		Manufacturer Specification[a]	AMTEC Verification
1	Main structure		
1.1	Overall dimensions, mm		
1.1.1	Length	1180	1400
1.1.2	Width	1035	1225
1.1.3	Height	1200	1190
1.2	Weight of machine without prime mover, kg	ND	NM[b]
2	Rated pelleting capacity, kg/h	204	452.75
3	Hopper		
3.1	Material	Stainless steel (1.2mm, #304)	303/304SS[c]
3.2	Dimensions, mm		
3.2.1	Diameter	320	305
3.2.2	Height	ND	175
3.3	Height from the ground, mm	1200	1190
3.4	Angle of inclination, °	ND	48
3.5	Dimension of inlet opening, D, mm	90	85
4	Feeder	NA	NA
5	Electric motor		
5.1	Type	Induction	Induction
5.2	Brand	MINDONG	MINDONG
5.3	Model	YC132SB-4	YC132SB-4
5.4	Serial number	1904003	1904003
5.5	Make or manufacturer	China	China
5.6	Rated power, kW	3.7	3.7[a]
5.7	Rated speed, rpm	1740	1740[a]
5.8	Electric service required	Single-phase	Single-phase
5.9	Voltage, V	220	220[a]
5.10	Current, A	30	30[a]
5.11	Frequency, Hz	60	60[a]

[a] The data were based on specifications sheet provided by the test applicant.
[b] Determined using BRUKER S1 TITAN 500 X-ray Fluorescence Analyzer. See Table 1 for elemental composition.
Note: NA – Not Applicable (The part is unavailable in the machine), ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer Specification[a]	AMTEC Verification
6	Conditioner	NA	NA
7	Pelleting chamber		
7.1	Type	ND	Die-rotating flat die pellet mill
7.2	Dimension, D × H, mm	ND	255 × 121.2
7.3	Roller		
7.3.1	Number of rollers	2	2
7.3.2	Material	Stainless steel	303/304SS[b]
7.3.3	Inner diameter, mm	32	38.08
7.3.4	Overall diameter, mm	ND	82.48
7.3.5	Width, mm	ND	77.38
7.3.6	Pattern or configuration	ND	Straight open-end corrugated roller
7.4	Barrel and screw	NA	NA
7.5	Die		
7.5.1	Material	Stainless steel (#304)	303/304SS[b]
7.5.2	Die diameter, mm	248	260
7.5.3	Overall width of ring die, mm	NA	NA
7.5.4	Working width of ring die, mm	NA	NA
7.5.5	Number of holes	ND	372
7.5.6	Diameter of holes, mm	ND	3.92
7.5.7	Open area, %	ND	8.46
7.5.8	Overall length, mm	ND	12.13
7.5.9	Effective length, mm	ND	6.64
7.5.10	L/D ratio	ND	1.69
7.5.11	Relief type	ND	Non-variable relief die
8	Discharge chute		
8.1	Material	Stainless steel (1.2mm, #304)	303/304SS[b]
8.2	Dimensions of opening, L × W, mm	450 × 103	450 × 100
8.3	Height from the ground, mm	475	475

[a] The data were based on specifications sheet provided by the test applicant.
[b] Determined using BRUKER S1 TITAN 500 X-ray Fluorescence Analyzer. See Table 1 for elemental composition.
Note: NA – Not Applicable (The machine is Flat die pellet mill, Die-rotating type), ND – No Data

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specifications[a]	AMTEC Verification
9	Frame		
9.1	Material	ND	Iron/CarbonSteel[b]
9.2	Overall dimensions, L × W, mm	ND	1035 × 1180
9.3	Height, mm	ND	80
10	Power transmission system		
10.1	Prime mover to pelleting assembly	V-belt and pulley	V-belt and pulley
10.1.1	Prime mover[c]	128 × 2B × 38	128.38 × 2B × 37.28
10.1.2	Pelleting assembly[c]	356 × 2B × 28	150.72 × 2B × 25.50
10.1.3	Belt size	B-51	B-51
10.2	Others	ND	a.) All moving or rotating parts are provided with covers. b.) On/off buttons

[a] The data were based on specifications sheet provided by the test applicant.

[b] Determined using BRUKER S1 TITAN 500 X-ray Fluorescence Analyzer. See Table 1 for elemental composition.

[c] Pulley diameter, mm × number of belt × shaft diameter, mm

Note: ND – No Data

RESULTS

Item		Data	
1	Conditions of test material		
1.1	Feed type	Feed Mix	
1.2	Source	Nabunturan, Davao de Oro	
1.3	Moisture content of feed mix, $\%_{wb}$	30.54	
2	Performance test		
2.1	Weight of input, kg	94.0[a]	
2.2	Weight of output, kg	91.4	
2.3	Moisture content of output, $\%_{wb}$	31.39	
2.4	Pelleting capacity, kg/h	452.75	
2.5	Pelleting recovery, %	96.88	
2.6	Pelleting efficiency, %	92.22	
2.7	Pellet diameter, mm	4.32	
2.7.1	Standard deviation, mm	0.097	
2.7.2	Coefficient of variation, %	2.24	
2.8	Speed of components, rpm	Without load	With load
2.8.1	Prime mover	1808	1769
2.8.2	Pelleting assembly	230	224
2.8.3	Reducer input	1506	1483
2.8.4	Reducer output	468	460
2.9	Noise at operator's ear level, dB(A)	Without load	With load
2.9.1	Feeding operator	94.7	86.1
2.9.2	Bagger	95.4	86.2
2.10	Power consumption	Without load	With load
2.10.1	Line voltage, V	233.8	227.9
2.10.2	Load current, A	8.4	25.9
2.10.3	Input power, kW	1.98	5.80

[a] Water was added on the input feed mix before the operation.

RESULTS (Cont'n)

Table 1. Elemental composition of the different components of the DAVAO TECHNOCRAFT GMPMVT1024 Feed Pellet Mill.

Element	Percent Composition[a]					
	Input hopper	Roller	Die	Discharge chute	Pelleting chamber	Frame
Titanium	0.030	0.024	<LOD	0.027	-	-
Vanadium	0.135	0.059	0.092	0.121	0.015	0.016
Chromium	18.191	19.605	18.383	18.154	<LOD	<LOD
Manganese	1.053	1.146	1.351	1.011	0.491	0.213
Iron	72.139	69.184	71.593	72.151	99.288	99.615
Cobalt	0.413	0.412	0.485	0.433	0.157	0.085
Nickel	7.974	9.379	7.953	8.046	0.013	0.010
Copper	0.035	0.121	0.063	0.037	0.025	0.029
Niobium	<LOD	0.016	0.004	<LOD	<LOD	<LOD
Molybdenum	0.013	0.044	0.038	0.012	0.011	0.014
Tin	<LOD	<LOD	0.025	<LOD	<LOD	<LOD
Tungsten	<LOD	<LOD	<LOD	<LOD	<LOD	<LOD
Lead	0.023	0.017	0.022	0.019	0.048	0.033
Zirconium	<LOD	<LOD	<LOD	<LOD	-	-
Tantalum	<LOD	<LOD	<LOD	0.012	-	-
Grade	**303/304SS**	**303/304SS**	**303/304SS**	**303/304SS**	**1000 Series LAS**	**Iron/ Carbon Steel**

[a] Determined using the BRUKER S1 TITAN 500 X-ray Fluorescence Analyzer.
Note: LOD – Limit of Detection

OBSERVATIONS

Figure 2. Quality of output.

Figure 3. Nameplate of the DAVAO TECHNOCRAFT GMPMVT1024 Feed Pellet Mill.

OBSERVATIONS (Cont'n)

Figure 4. Prime mover nameplate of the DAVAO TECHNOCRAFT GMPMVT1024 Feed Pellet Mill.

Figure 5. Labelling on the DAVAO TECHNOCRAFT GMPMVT1024 Feed Pellet Mill.

OBSERVATIONS (Cont'n)

Figure 6. Warning notices and nameplate of the prime mover of the DAVAO TECHNOCRAFT GMPMVT1024 Feed Pellet Mill.

Figure 7. Extra pellet dies provided by the applicant for DAVAO TECHNOCRAFT GMPMVT1024 Feed Pellet Mill.

Tested by:

GLORY YSTER M. REGINIO
Test Engineer

Verified by:

RHEY MARC C. ALBALOS
Test Engineer

Approved for Release:

ARTHUR L. FAJARDO	11 DEC 2024
AMTEC Director	Date

REMINDER

This test report should not be used for advertisement or for promotion of the product but only for evaluation and verification purposes and should not be reproduced in whole or in part without the permission of AMTEC.

For inquiries regarding the test report, you can visit us at our office, contact us at +63 921 400 7137 or +63 917 704 0922, send us a message at amtec.uplb@up.edu.ph or visit our official website at www.amtec.ceat.uplb.edu.ph.